“What was where?” Visual short-term memory binding test - Insight46 phase 1

Kirsty Lu | 24/07/2019

Document details

Categories of variables	“What was where?” Visual short-term memory binding test
Summary of work undertaken	1. Data collected, processed and cleaned as described below. 2. Summary outcomes calculated as described below.
Names of source variables	For, descriptions, of, what, these, variables, mean, see, the, labels, in, the, dta, file, and, the, papers, by, Pertzov, et, al, referenced, below, id, (participant’s, ID, number), Bload, Bdelay, Bid, Blo, Bnic, Bswap, Bcorrlswap, Bcorriswap, Bcorrbswap
Output variables	binding_comp_i46p1, binding_wn_i46p1, binding_ontrials_i46p1, binding_oncorrect_i46p1, binding_oid_i46p1, binding_olo_i46p1, binding_onic_i46p1, binding_onswap_i46p1, binding_ncorrect3_i46p1, binding_oswap_i46p1
Papers using these variables	Characterising cognition in later life and its relationship with biomarkers of Alzheimer's disease: a study of members of the National Survey of Health and Development (the British 1946 birth cohort). PhD thesis (Kirsty Lu) **It is recommended that potential users of these datasets discuss their analyses with Kirsty Lu.**

Background

The “What was where?” test was designed by Dr Yoni Pertzov (Hebrew University of Jerusalem) and Professor Masud Husain (University of Oxford). A brief description has been published in the Insight 46 protocol paper: 1. Lane et al. (2017) Study protocol: Insight 46 – a neuroscience sub-study of the MRC National Survey of Health and Development, BMC Neurology, 17:75, doi: 10.1186/s12883-017-0846-x See also papers by Pertzov et al. e.g. 1. Pertzov et al (2015) Effects of healthy ageing on precision and binding of object location in visual short term memory. Psychology and Aging, 30(1), 26–35 2. Liang, Pertzov et al (2016) Visual short-term memory binding deficit in familial Alzheimer's disease. Cortex, 78, 150-164

In Insight 46, the participant was seated in front of a DELL Optiplex 9030 all-in-one touchscreen computer. The dimensions of the screen were 51.2 x 28.7 cm and the approximate distance from the subject’s eyes to the centre of the screen was 58 cm. In each trial, one or three fractals were displayed on the screen in random locations, presented on a black background. Participants were asked to look at the fractals and to try to remember their identities and locations. (See Figures in the above papers.) 1-fractal trials are referred to as ‘low load’ and 3-fractal trials are referred to as ‘high load’. The low load trials were displayed for 1 second whereas the high load trials were displayed for 3 seconds to allow time for encoding. This was followed by a blank screen for either a short or long delay (1 or 4 seconds), and then a test array appeared in which two fractals were displayed along the vertical meridian. One of these fractals had appeared in the memory array on the previous screen (the target) and the other was a foil or distractor. Participants were instructed to touch the fractal that they remembered seeing and drag it to the location where they think it was originally presented. There was no time-limit for reporting the location – the tester pressed the space bar to initiate the next trial when the participant was ready. A shortened version of the task was designed specifically for Insight 46 by Dr Pertzov, containing 24 trials: 4 low load and 20 high load. Within each load condition, trials were equally likely to have a short or long delay. Therefore, there were four possible combinations of load and delay (2 x low load with short delay; 2 x low load with long delay; 10 x high load with short delay; 10 x high load with long delay). The experiment was preceded by 4 practice trials – one of each of the load x delay combinations, and the tester ensured that the participant understood the task before continuing. The practice trials are not included in this dataset. The locations of the fractals were generated in a pseudo-randomised manner by a MATLAB script (MathWorks, Inc). The script imposed the following restrictions which are necessary to allow analysis of localisation error which is a key outcome of this task: fractals were always at least 9o away from each other to avoid crowding and to ensure that there was a clear zone of 4.5o around each fractal which is necessary for the calculation of swap errors (see below), and fractals were at least 6.5o from the centre of the screen and 3.9o from the edges. The 24 trials were the same for all participants (i.e. the same fractals were presented in the same locations) but the trials were presented in a random order for each participant. For details of the outcome measures, see the Pertzov papers cited above.

Data processing and cleaning

Each participant's raw data file was processed using a MATLAB script written by Dr Pertzov, which generated a score for each outcome variable in each of the four conditions (low load with short delay; low load with long delay; high load with short delay; high load with long delay). Data were imported into Stata, creating one dataset with 4 rows per participant (one for each of the four conditions).
1. Data were checked to identify if all data-points were plausible. It was noted that six participants had one trial where the software did not record whether they selected the correct or incorrect fractal. I corresponded with Dr Pertzov, and he concluded that this was likely to be caused by the participant touching the screen exactly midway between the two fractals. These six individual trials were excluded and the mean scores for the relevant conditions were recalculated. 2. In light of this, a new variable was calculated to indicate how many usable trials each participant had within each condition (Bntrials). 3. For Kirsty Lu's statistical model, it was useful to transform the proportion of correct identifications (Bid) into the number of correct identifications (Bncorrect), which was done by multiplying Bid x Bntrials. 4. It was also useful to transform the proportion of swap errors within each condition (Bswap) into the number of swap errors (Bnswap), which was done by multiplying Bswap x Bncorrect. This dataset was saved as Insight46_binding_fulldata_cleaned_final_20190724
Note that it contains 486 participants, as the remaining 16 did not complete the binding task. For details of participants who did not have useable data – and reasons why – see the summary dataset described below.

A second dataset was created which contains one row per participant, summarising their performance across all 24 trials of the experiment. The following variables were calculated: 1. The total number of valid trials for each participant (Bontrials) was calculated by summing their four values of Bntrials. Note that this variable equals 24 for all participants except the 6 people who had an invalid trial excluded, as explained above. 2. The total number of correct responses for each participant (Boncorrect) was calculated by summing their four values of Bncorrect. 3. Each participant's overall proportion of correct responses (Boid) was calculated by dividing Boncorrect / Bontrials 4. The mean gross localisation error (Bolo) was calculated for each participant by finding the average of Blo across the four conditions, weighed according to the number of correct responses made within each condition. 5. The mean 'nearest item control' localisation error (Bonic) was calculated for each participant by finding the average of Bnic across the four conditions, weighed according to the number of correct responses made within each condition. 6. The total number of swap errors for each participant (Bonswap) was calculated by summing their two values of Bnswap across the two high-load (3-item) conditions (note that swap errors cannot be made on the low load conditions). 7. In order to transform this into the proportion of swap errors, the number of correct responses for each participant across the two high-load conditions was first calculated (Bncorrect3). The proportion of swap errors (Boswap) was then found by dividing the number of swap errors by this number (Bnswap / Bncorrect3). This dataset was saved as Insight46_binding_summaryoutcomes_cleaned_final_20190724 Note that it contains 502 participants, of whom 486 completed the binding task. The variable Bcomp identifies which participants completed the binding task. The variable Bwn gives the reasons for any missing data. These two variables were imported from XNAT.

Variables in this document

9 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Visual short-term memory [60428]		

Variable	Label	How it was found
binding_comp_i46p1	Binding: Was the task completed? - Insight46 phase 1	topsheet field
binding_ncorrect3_i46p1	Binding: total number of correct trials in the 3-item condition - Insight46 phase 1	topsheet field
binding_oid_i46p1	Binding: proportion of correct responses, identification rate across all trials - Insight46 phase 1	topsheet field
binding_olo_i46p1	Binding: overall mean localisation error across all correct trials - Insight46 phase 1	topsheet field
binding_oncorrect_i46p1	Binding: total number of correct responses for each participant - Insight46 phase 1	topsheet field
binding_onic_i46p1	Binding: mean nearest item control across all correct trials - Insight46 phase 1	topsheet field
binding_onswap_i46p1	Binding: total number of swap errors - Insight46 phase 1	topsheet field
binding_ontrials_i46p1	Binding: total number of valid trials - Insight46 phase 1	topsheet field
binding_oswap_i46p1	Binding: overall absolute swap error rate - Insight46 phase 1	topsheet field